signal consulting, inc.

RE: Connector Ground Pin Patterns

On a recent flight I had the opportunity to work out an example of all the equations needed to estimate crosstalk on a board-to-board connector (see attached file). If this spreadsheet is of use to you, you are welcome to have it. I am glad that I put it together, because I get a lot of questions about this subject. You will of course want to adjust my guesses as to the physical dimensions of the connector, and the exact arrangement of ground pins. If I were you, I would have MathCad read in the ground and signal positions from a text file if they are irregular in nature and not easily generated from a simple equation.

If you do not have a copy of the MathCad application, you may want to get one. It's only about $100 from a company called Mathsoft (check the 800 number directory for a listing). Mathsoft takes credit cards and ships overnight. You will need version 6.0 for Windows.

I tried two arrangements of ground pins, both involving the same connector geometry. The connector in both cases was a four-row connector; with 1/2 inch total pin length (going up from the board, across to the other connector, and back down again - we have to guess an average effective value for this parameter). I assumed a 0.05" pin spacing on each row. The drive currents were set to represent dI/dt= 4 volts into 50 ohms at 1 ns rise/fall. You can scale your results to other rise/fall times or other signal sizes.

Also, because the connector length parameter is a *pretty gross assumption*, you will want to make a measurement of the crosstalk on some sample signal pair and correlate it with these plots. A scaling factor will probably be needed to correct my result closer to reality. I believe that this method, while it may not be accurate for calculating exact results, is *fairly* accurate for evaluating the *differences* between connector ground patterns.

The two ground configurations are illustrated in figures 1 and 2.

Figure 3 shows the total accumulated crosstalk on each pin as a function of pin number for our two different configurations of ground pins. It assumes, for each pin, that all other pins happen to be changing in their worst case pattern at the time we make our crosstalk measurement.. Circles represent crosstalk measurements made using the ground configuration in Figure 1, crosses represent crosstalk measurements made using the ground configuration in Figure 2.

CONCLUSIONS:

- Overall, the interspersed-ground connector worked much better.
- With 1 ns rise/fall times and the ground configuration of Figure 2 you will get about 1 volt of crosstalk. In other words, the connector doesn't work very well with fast signals.
- Pins at the ends of the rows are often 10-20% different from more central pins.
- Pins 7, 11, 15... are slightly preferred, as they are each near two grounds and have few nearby disturbers.

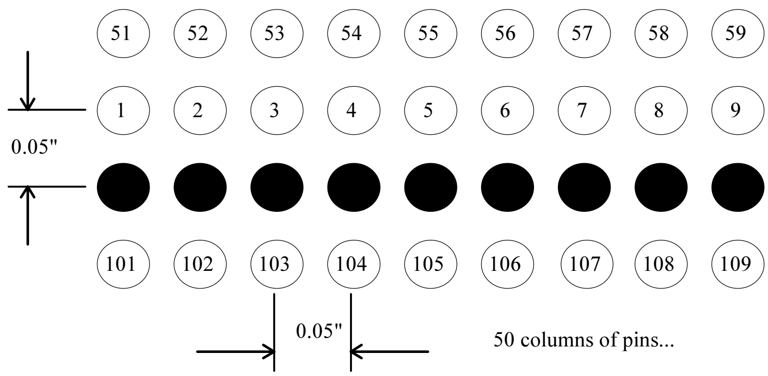


Figure 1—Single Row of Ground Pins

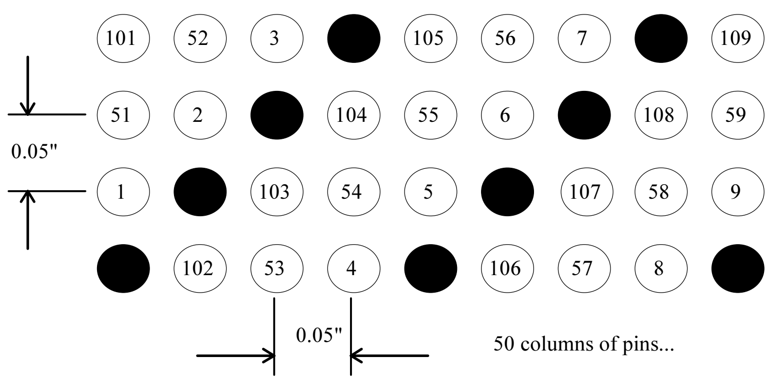


Figure 2—Ground Pins Interspersed with Signals

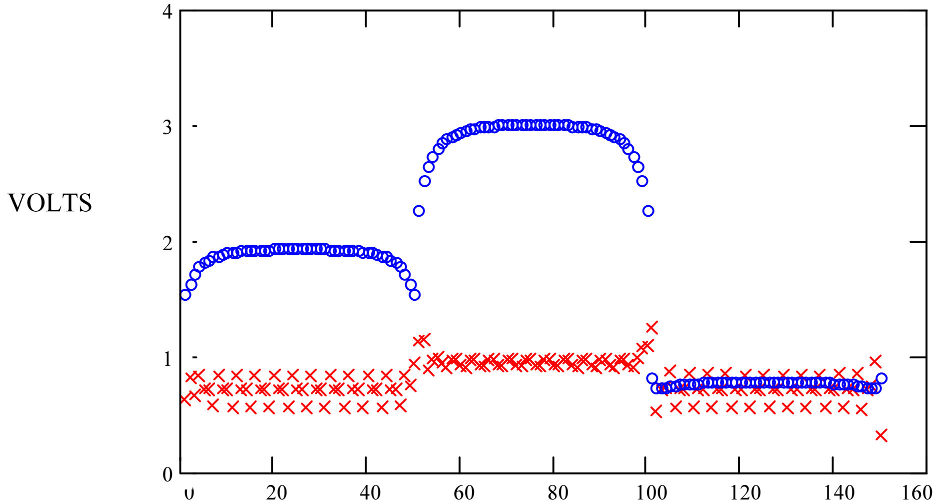


Figure 3—Total Worst Case Predicted Crosstalk by Pin Number for Two Different Ground Configurations

see also MathCad file: gndpins.xmcd

This is an experimental spreadsheet used to gauge the impact of various patterns of ground/power pins in a high-speed digital connector. See MSWORD 6.0 file gndpins.doc .

Gndpins.mcd -- crosstalk in connectors

Magnetic field intensity at a point x, given current flow of 1 amp in a wire located at position p. All positions in a 2-dimensional plane. Positions on the plane are represented by complex quantities (this results in more convenient notation than if I had used little 2-element vectors to represent each quantity).

Magnetic permeability of free space $\mu 0 := 4 \cdot \pi \cdot 10^{-7}$

Wire radius (m) $R := 0.01 \cdot .0254$

$$B(x,p) := \mathrm{if}\left(|x-p| > R, \frac{\mu 0}{2 \cdot \pi \cdot |x-p|}, 0\right)$$

Length of connector pins

Wire length (m) $L := 0.5 \cdot .0254$

Elemental flux induced at point v, located on flux integration path between points X and Y, due to durrent flow of 1 amp at position p.

Dot product operator applied to our complex position variables

$$\mathrm{dot}(a,b) := \mathrm{Re}\left(a \cdot \overline{b}\right)$$

$$d\phi(v,x,y,p) := B(v,p) \cdot \frac{\mathrm{dot}[(y-x),(p-v)]}{|y-x| \cdot |p-v|}$$

Integral of flux between wires located at points X and Y, due to flow of 1 amp at position p

$$\phi(x,y,p) := \int_x^y d\phi(v,x,y,p) \cdot L \, dv$$

Where the integration variable, v, traces a path from x to y

Change the flux integration path to first circumscrbe a circle of constant radius about the current source, starting at point x andending at a location nearest to point y. Now complete the integral, moving radially out towards point y. On the circumscribed path, the dot product is always zero, so we get no result. On the radial path, the dot product is always unity. Furthermore, along the radial path the integrand is a function of 1/R, which integrates to log(R).

$$\phi(x,y,p) := \frac{\mu 0}{2\cdot\pi}\cdot\ln\left(\frac{\text{if}(|y-p| > R, |y-p|, R)}{\text{if}(|x-p| > R, |x-p|, R)}\right)\cdot L$$

Define wire positions. M and N need not necessarily be equal.

Standard pin spacing (m) $\quad D := 0.05\cdot 0.0254$

$N := 50 \qquad n := 0, 1 .. N-1$

Ground wires, row 1 $\quad g_n := n + j$

Where the position "g" has an imaginary "j" (vertical) component and a real (horizontal) component.

Signal wires, row 2 $\quad p1_n := n + 2j$

row 3 $\quad p2_n := n + 3j$

row 0 $\quad p3_n := n + 0j$

This arrangement has all the grounds on row 1. Change the equations to scatter the ground positions.

$M := N\cdot 3 \qquad m := 0, 1 .. M-1$

$\text{append}(a,b) := \text{augment}\left(a^T, b^T\right)^T$

$p := \text{append}(p1, \text{append}(p2, p3))$

$\text{fixpos}(x) := D\cdot(\text{Re}(x) + j\cdot\text{mod}(\text{Im}(x), 4))$

$p := \overrightarrow{\text{fixpos}(p)}$

$g := \overrightarrow{\text{fixpos}(g)}$

Matrix showing flux induced in various ground loops, due to flow of 1 amp in wire located at position p

$\text{nLessOne} := 0, 1 .. N-2$

$B_{\text{nLessOne}, m} := \phi\left(g_{\text{nLessOne}}, g_{\text{nLessOne}+1}, p_m\right)$

Matrix relating flux in each ground loop to flow of current in other ground wires.

$A_{\text{nLessOne}, n} := \phi\left(g_{\text{nLessOne}}, g_{\text{nLessOne}+1}, g_n\right)$

Column <m> of matrix B shows how signal currents affect the flux in each ground loop. The matrix A shows how ground currents affect the flux in each ground loop. For each possible signal current (that is, for each column of B), the pattern of ground currents must be such that the total flux through each ground loop is zero. Arranging the ground current solutions as columns of a matrix G, we get:

$A\cdot G^{\langle m\rangle} + B^{\langle m\rangle} := 0$ $\qquad$ - OR - $\qquad$ $A\cdot G + B := 0$

The above constraints are insufficient to completely solve our problem. The matrix A is of size (N-1)xN, because while there are N ground currents, there are only (N-1) ground current loop constraints. We need an Nth constraint to solve the problem. The last constraint is found by forcing the sum of all ground currents to equal the signal current (that is, all curent going out must find a way back along the grounds). We will add an Nth row to matrices A and B to effect this constraint.

Bottom row of matrix represents our last constraint, which says that the sum of all currents has to equal zero.

Sum of ground currents and source current equals zero

$$A_{N-1,n} := 1$$

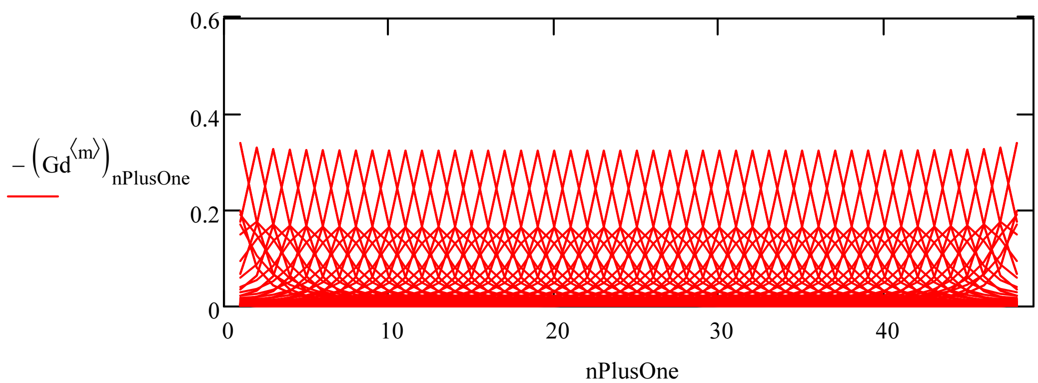
$$B_{N-1,m} := 1$$

Solve for vectors of ground currents. Each column represents ground excitation by a different agresso source.

$$G := -\left(A^{-1} \cdot B\right)$$

Plot out calculated versus ideal distribution

$nPlusOne := 0, 1 .. N$ $\quad Gd := G$ $\quad Gd_{N,m} := 0$ $\quad g_N := 0$



Compute crosstalk voltage received between wires positioned at any points x and y in space, as a result of current flowing on wire m, with consideration given to currents induced on all ground wires.

Aggressor current dI/dt on each signal

$$dIdt_m := \frac{4}{50} \cdot \frac{1}{10^{-9}}$$

Note: these could be different for different net classes

$$voltage(x,y,m) := \left[\phi\left(x,y,p_m\right) + \sum_n \left(\phi\left(x,y,g_n\right) \cdot G_{n,m}\right)\right] \cdot dIdt_m$$

Matrix of output crosstalk voltages, as a function of signal currents

$$k := 0, 1 .. M - 1$$

$$V_{k,m} := \mathrm{voltage}\left(p_k, g_0, m\right)$$

Vectorize the above equation

How signal currents affect other signal voltages

$$C_{k,m} := \phi\left(p_k, g_0, p_m\right)$$

How ground currents affect signal voltages

$$D_{k,n} := \phi\left(p_k, g_0, g_n\right)$$

$$V := C + D \cdot G$$

Scale by dIdt on each signal wire

$$V_{k,m} := V_{k,m} \cdot dIdt_m$$

Zero out self-crosstalk terms

$$V_{m,m} := 0$$

Evaluate worst case crosstalk on a line-by-line basis

$$V_{k,m} := \left|V_{k,m}\right|$$

$$V := V^T$$

$$XTLK_m := \sum V^{\langle m \rangle}$$

WRITEPRN("straight.prn") := XTLK

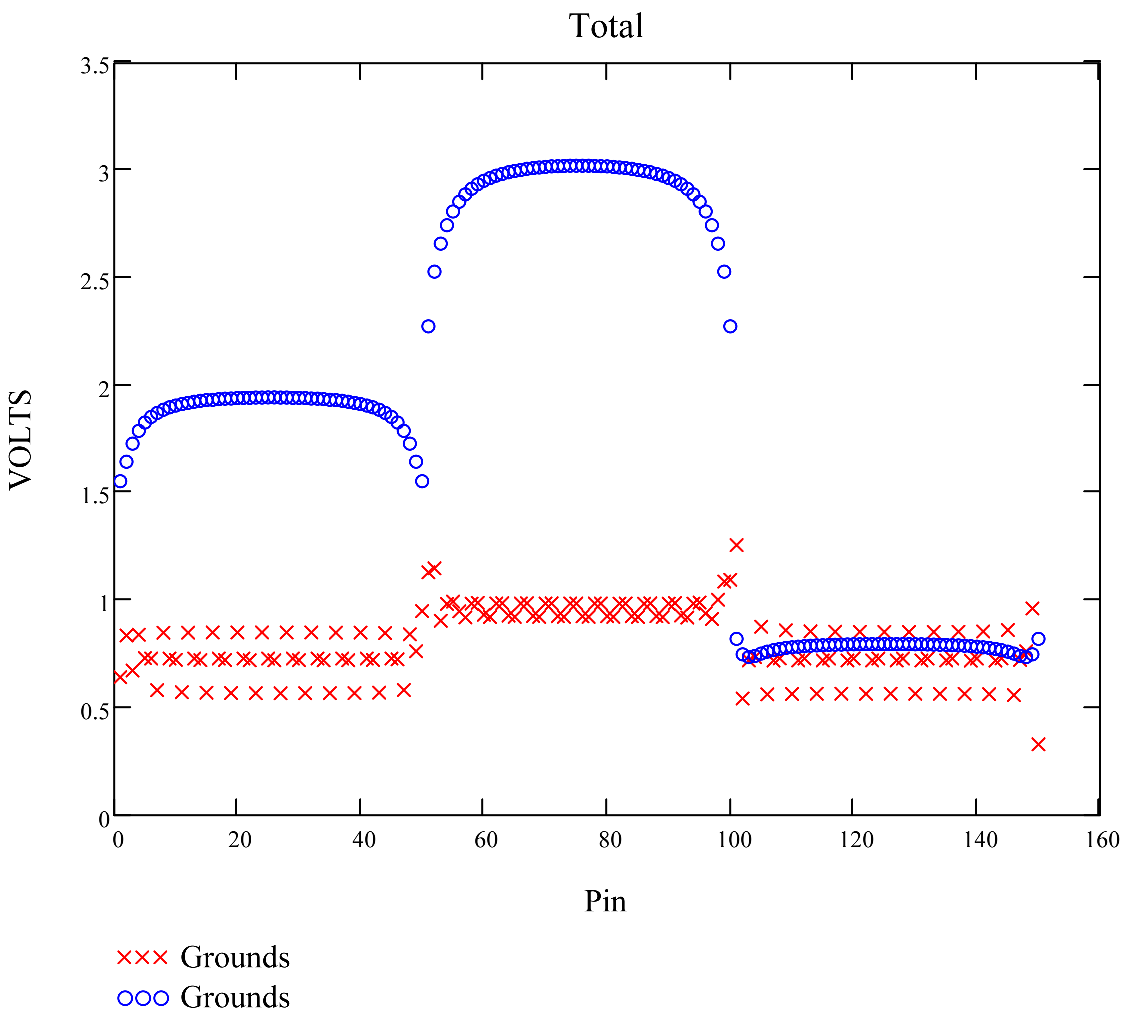
WRITEPRN("scatter.prn") := XTLK

I have pre-computed and stored the two configurations documented in gndpins.doc.

XSTRAIGHT := READPRN("straight.prn")

XSCATTER := READPRN("scatter.prn")

The graph below displays these two pre-computed solutions.


Total
VOLTS
3.5
3
2.5
2
1.5
1
0.5
0
0
20
40
60
80
100
120
140
160
Pin
××× Grounds
ooo Grounds